

Web Pages as 3D Proxies in VR: A Design Framing and Preliminary Exploration

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Abstract

When users work with and manage multiple pages in Virtual Reality (VR), interaction still largely follows the same 2D tab and window patterns as traditional browsers. We propose a design framing for VR web page management that represents each page as a virtual 3D proxy inspired by familiar real-world objects, endows these proxies with physical properties, and uses them to create a 3D workspace where pages can be organized, handled, and stored through everyday physical interactions. To explore this framing, we built a proof-of-concept prototype with representative features that instantiate these ideas. We then conducted a preliminary evaluation suggesting a moderately positive overall attitude toward usability and that proxy-based page management can feel intuitive and engaging. Finally, we outline future directions for proxy-based VR web page management, including additional interactions and system features.

CCS Concepts

• **Human-centered computing** → **Virtual reality**.

Keywords

web page management, virtual reality, 3D workspace, object-based interaction

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1 Introduction

Virtual Reality (VR) immerses users in interactive three-dimensional environments through head-mounted displays and motion-tracked input [7], and has increasingly been explored as a platform for moving beyond traditional 2D screen-based workspaces [11]. A key direction in VR workspace design is to make interaction feel more natural and efficient by grounding input in users' embodied skills rather than relying solely on conventional 2D GUI operations

[3, 12]. Object-based and physical interaction techniques introduce virtual proxies of familiar tools and artifacts so users can grab, reposition, rotate, and manipulate interface elements with motions that mirror everyday object use [6]. By mapping digital actions onto stable, object-centric representations in space, virtual proxies can offer clearer affordances, reduce abstract command overhead, and support more direct organization of information in immersive work settings [2, 10]. Prior work suggests that VR workspaces can benefit from treating interface elements as manipulable objects, allowing users to utilize physical intuition and spatial memory when working in immersive environments [11, 21].

Despite these advantages, most current VR web applications still manage web pages as 2D windows or tabs and rely on traditional screen-based operations, even when those pages are displayed within an immersive 3D workspace. Prior systems have explored adding spatial structure to page or file organization. For example, early 3D workspaces that place pages on virtual desks and store them in metaphorical containers (e.g., books or shelves) [5], desktop environments supporting physics-inspired organization such as piling, stacking, and spatial grouping [1], and VR visualizations that arrange pages or histories in 3D tree-like structures [9]. However, relatively few systems treat web pages as tangible objects that can be organized in space the way people handle everyday items. Early work explored organizing web pages spatially on 2D displays to take advantage of spatial memory [19]. Recent immersive analytics and virtual workspace research highlights how users exploit spatial memory and externalization by arranging many document-like items around them [14, 16], and how space-saving physical organization behaviors such as stacking and grouping emerge when managing collections in immersive space [17, 18]. Motivated by these observations and prior mixed reality systems for object-based physical interaction [8, 13], we represent web pages as virtual proxies of familiar real-world objects to support more physical and spatial tab management in VR.

Our design framing creates 3D proxies of real-world objects for web tabs in VR that reflect page characteristics, such as content type. The tab represented by the proxy can therefore be assigned physical properties to convey states such as importance or activeness, and can be freely manipulated, used, and stored within a 3D workspace through physical interaction. We instantiated this framing in a proof-of-concept prototype that demonstrates representative features, and conducted a preliminary evaluation to probe usability and the perceived value of this direction. Our contributions are (1) a design framing for managing web pages in VR through 3D proxies that mirror everyday object management, and (2) an initial exploration of the framing that demonstrates its potential and informs

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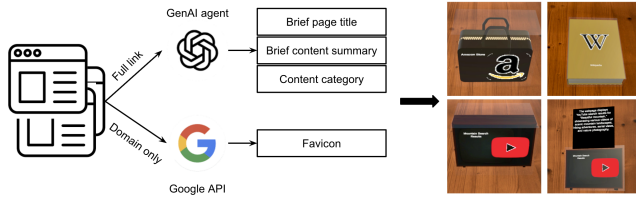


Figure 1: Pipeline of proxy generation in our prototype. Web pages become 3D proxies of real-world objects that represent different content categories, with a content summary available on demand.

future designs. Section 2 details the framing and prototype, and Section 3 describes the preliminary user study design and results.

2 Web Page Management with Virtual Proxies

In this section, we present our design framing for VR web page management, describing how tabs can be represented as manipulable 3D proxies and organized in a spatial workspace. We present the framing as three sequential stages that build on each other: (1) representing web pages as 3D proxies, (2) endowing proxies with physical properties, and (3) using them as building blocks for a virtual workspace. For each stage in the framing, we summarize the idea, illustrate it with representative prototype features, and note opportunities for future extensions.

2.1 Proxy Representation for Web Pages

Representing each web tab as an object-like 3D proxy enables physical manipulation and improves awareness when managing many pages in VR. The proxy retains the core information of a traditional tab (e.g., title, favicon, URL) but avoids the readability limits of crowded tab strips. It can also display additional and glanceable cues beyond what a 2D tab typically provides. In our prototype pipeline (Figure 1), when a user minimizes a web view, an AI agent (GPT-4.1) generates a short title shown on the proxy and a hidden brief summary that the user can check at any time. The agent also assigns the page to a preset category (e.g., video, knowledge, shopping) and selects a corresponding real-world-inspired proxy form (e.g., TV, book, bag). These cues help users assess a page without reopening it and support later organization in the 3D workspace.

Future designs could make proxy generation more customizable and expressive. Users could choose what metadata is shown (e.g., page language, last-updated time) and how proxy appearance encodes content (e.g., food pages represented by cuisine-specific objects) based on their needs under different circumstances. Proxy creation could also be rule-based or using user-authored templates rather than fully AI-driven, improving consistency and giving users more control over how pages are represented.

2.2 Endowing Physical Properties for Web Page Proxies

Instead of being stationary props in the virtual environment, the virtual proxies that represent web tabs can be manipulated much like real-world objects. In VR, we can assign them physical properties to



Figure 2: Illustration of example physical properties of proxies in our prototype. (a) Proxies can be resized, stacked, and placed in the virtual environment. (b) Inactive proxies gradually become frozen.

enable different types of manipulation. In our prototype (Figure 2), proxies can be rotated, thrown, resized, and freely placed or stacked in the 3D environment. These manipulations may be more suitable and intuitive in VR, where spatial interaction is central, compared to managing web pages through traditional 2D tabs. Physical properties can also communicate page state. For example, if a proxy has not been used for a while, it gradually “freezes,” signaling that the page is inactive or likely no longer needed.

Because real-world objects have various physical properties, this step opens rich possibilities for future implementations, both for conveying more information and enabling new functions. Proxies could also use familiar physical metaphors to show functional states. For example, “pinned” proxies that resist movement, “glowing” proxies that draw attention, or “compressed” bundles that expand on demand. Because these properties and metaphors come from everyday experience, they may support natural interaction with a low learning cost. Physical properties could also enable novel and meaningful functions related to page management in 3D. Future proxies could vary in weight or friction, snap together with magnet-like behavior, or form elastic groups that stay linked while still allowing rearrangement.

2.3 Virtual Workspaces with Web Page Proxies

Traditional web browsers create a 2D GUI workspace for managing multiple web pages. In VR, simply carrying over this design misses the opportunity of 3D spatial interaction. Under our framing, the virtual proxies from the previous steps become the building blocks of a VR workspace: when working with many web pages, users can arrange and handle them in space as if managing physical objects in a real workspace, with added virtual-only functions. We expect this supports a more realistic and intuitive interaction experience that better matches VR, especially when users manage multiple pages concurrently. In our prototype workspace shown in Figure 3a, users interact with several screens that display web views. When a user minimizes a page, a proxy is generated on a stand near the table. To reopen a page on a screen, the user places the proxy onto that screen’s dedicated projection zone on the table. To temporarily store proxies, the user can grab a basket with adjustable size from a nearby shelf and place proxies inside it. Users can also resize proxies to indicate importance or simply make them easier to find. If a proxy is no longer needed, the user can throw it into a trash can underneath the table to destroy it.

If the workspace becomes crowded, the basket can be transformed into a lockbox (shown in Figure 3b) that shows the favicons of the proxies inside, records its creation timestamp, and can be



Figure 3: (a) Virtual workspace in our prototype (details in Section 2.3). (b) Proxies can be stored in a lockbox with a user-defined name and creation timestamp (left) and favicons of the contained web pages (middle). Lockboxes can be placed in a cabinet for long-term storage (right).

named by the user. The user can later open the lockbox or place it into long-term storage in a cabinet. This workflow aligns with prior findings that users employ staging or buffer areas to temporarily park and triage pages when working with multiple virtual pages [15]. The cabinet also supports longer-term storage, resembling bookmarking tabs for later use in traditional browsers. As shown in Figure 3b, lockboxes can either be placed freely for manual retrieval (right side of the cabinet), or thrown into a glowing region for the cabinet to automatically organize them in a dedicated area (left side of the cabinet), which requires the user to use the center UI to select and retrieve the lockboxes they want. As shown in Figure 3a, we also added a bell representing the home page and three spheres representing favorite pages on the table, which users can tap to open the corresponding pages.

Future designs can further enrich how the workspace supports proxy use by providing dedicated areas and reusable structures for common activities. For example, the workspace could include a “review” area to quickly scan a set of pages and a “staging” area to hold pages needed soon. It could also offer optional structures such as racks, slots, or trays that help users lay out proxies as lanes (e.g., “to read,” “in progress,” “done”) and keep collections organized during multitasking. Building on this idea, workspaces can have template sets of proxy slots, holders, and projection zones that can be saved, duplicated, and restored for different activities (e.g., “writing,” “shopping,” “video watching”). Cross-workspace movement for collaboration can also be added. Proxies could be passed between workspaces as a way of handing off context, collecting materials, or staging items for later. In collaborative settings, passing proxies could become a lightweight way to share not just links, but also structured context (e.g., a grouped stack with annotations, or a curated kit of pages relevant to a task). The storage function shown

Table 1: List of open-ended questions used in the post-study interview.

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- Q1. What do you think about representing tabs as virtual proxies with different forms for different content?
- Q2. How did the interactions with the proxies and other tools feel?
- Q3. Compared with a traditional browser, how efficient was switching between tabs in this system?
- Q4. How did you like the tab organization and storage methods? Did they match how you manage tabs in a traditional browser?
- Q5. Did the proxies, containers, and boxes give enough information about the tabs they represented or contained?
- Q6. Which workspace features were most useful?
- Q7. Was the “freezing” effect helpful for showing that a page was no longer in use?
- Q8. Did you prefer automatic or manual storage? Why?
- Q9. What did you like most, and what should be improved?
-

in our prototype can be expanded as well. Future systems could support partial unpacking, multi-level organization (i.e., boxes within boxes), and enable faster search by generating a meta summary of the pages in each box for quick preview. They could also support “smart” behaviors, such as auto-suggested grouping based on usage patterns. In addition, proxies could generate history-aware cues (e.g., recency, frequency, unfinished status) that help users decide where to store them.

3 User Feedback

3.1 Study Design

To evaluate the usability of the system and identify potential areas for improvement, we conducted a small-scale evaluation approved by the institution’s Institutional Review Board with five participants (mean age = 23.6). Participants used our VR prototype on a Meta Quest 2 headset. After a training session introducing the core features, participants completed an individual task: planning a five-day trip to London from New York City. To encourage engagement with diverse proxy types and organizational behaviors, participants were prompted to consider multiple planning aspects (e.g., food, lodging, transportation, entertainment, and shopping). Participants were asked to save relevant web pages during the planning process and store the resulting proxies in the storage cabinet. We intentionally provided no constraints on how participants should search, organize, or store information, and imposed no time limit, aiming to observe self-directed strategies supported by the proxy concept. After completing the task, participants filled out the System Usability Scale (SUS) questionnaire [4] on a 5-point Likert scale. We also asked the open-ended questions in Table 1 to gather qualitative feedback on proxy representations, interactions, organization, and the overall workspace.

3.2 Preliminary Findings

We reverse-coded negatively worded SUS items and computed an overall usability score. Across five participants, the mean item rating was 3.76 (SD = 0.57) on a 5-point scale, indicating a moderately

positive usability attitude. All items averaged ≥ 3 (neutral or above). The lowest-rated items were Q5 (function integration) and Q8 (cumbersomeness), both with an average of 3, suggesting opportunities to better streamline workflows and simplify controls.

Open-ended questions suggested several themes. Participants noted that in traditional browsers, distinguishing many open tabs often relies on memory and repeated switching. In contrast, virtual proxies were described as more intuitive and informative, helping them recognize pages and locate targets quickly. Physical interactions such as throwing and resizing also felt engaging and natural. During the study, when many pages were open, proxies sometimes took up substantial space and made specific items harder to locate than clicking through 2D tabs. In these cases, containers were described as especially helpful. Suggested improvements included richer container and box summaries, plus functions such as auto-sorting placements or dynamically resizing proxies to manage space. For proxy storage, all participants liked the lockbox and cabinet metaphor and felt it provided sufficient information. Some compared it to bookmarks, which made it easy to understand. Four participants preferred automatic storage because it reduced effort. One participant noted that manual storage may become more useful as collections grow and customized organization is needed. The frozen-item effect was helpful for two participants, but one participant found it distracting and suggested using a longer delay before it activates. This is similar to a known tradeoff in skeuomorphic versus flat user interface design: adding physical-metaphor cues can help communicate interface state intuitively, but may also feel visually distracting [20]. Finally, one participant wanted more direct, real-time control over the AI prompt for generating the proxies.

4 Conclusion and Future Work

In this work, we introduced a design framing for VR web page management that moves beyond traditional 2D tab and window setups by representing each page as a virtual 3D proxy inspired by familiar real-world objects. By assigning proxies physical properties and using them as building blocks in a spatial workspace, this framing supports organizing, handling, and storing pages through physical-like interaction in VR. We instantiated the framing in a proof-of-concept prototype and conducted a preliminary formative study to gather early usability and perception feedback. Together, the prototype and study feedback suggest that proxy-based web page management can feel intuitive and engaging in VR, and that participants had positive expectations about its value for organizing and working with multiple web pages in immersive environments.

Our work has several limitations, including the early-stage and limited-scope nature of the proof-of-concept VR prototype and the small-scale user study with only a single exploratory task. Future work can further explore the design framing in Section 2 by implementing a more customizable and personalized proxy-generation pipeline, richer physical properties and workspace support features for interacting with and organizing multiple proxies, and collaboration across multiple workspaces. Prior approaches that use floating 2D virtual windows to exploit spatial advantages can face clutter as the number of windows grows, which may degrade performance and user experience [22]. Accordingly, an important direction for

future work is improving scalability to avoid proxies becoming cumbersome as the number of pages grows, a concern also reflected in our preliminary feedback. Future systems could reduce overhead through hierarchical representations (e.g., larger “category proxies” that break into individual page proxies on demand) and through automatic grouping or clustering based on page category or content, potentially enabled by AI. In addition, the virtual workspace could be extended, particularly in collaborative settings, to support the distribution of proxies across multiple locations or shared work areas. Finally, the framework may extend beyond VR to support applications in other mixed reality settings in future explorations.

Follow-up evaluations can directly compare proxy-based management with traditional browser-style approaches, incorporate objective performance metrics, and evaluate longer-term use through a longitudinal study. Evaluations can also explore the tradeoff between intuitive, fun physical interaction and efficiency and ease of use at scale, across contexts such as personal browsing versus office productivity, and across users with different levels of VR experience. As we contribute a design framing rather than a finished system, we hope it can be explored, extended, and assessed by the broader community.

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